



Generating 1.9 Million HTML Pages to Explain How the Cook County Assessor's Statistical Model Assessed Your Home

Nicole Jardine, Michael Wagner, Jean Cochrane



COOK COUNTY
**ASSESSOR'S
OFFICE**

+ shoutout to Dan Snow and Claire Boyd!

Data Science and Transparency at the Assessor's Office

The Data Team at the Cook County Assessor's Office is responsible for **the model**.

the model: a computer program that uses vast amounts of data about past home sales to estimate current values.

This model produces estimated property values for roughly 1 million homes in Cook County.

Our goals are to be **accurate, equitable,** and **transparent.**

The big transparency question: **How did the model assess my home?**



“How did the model assess my home?”

To estimate your home's value in a way that fairly and accurately follows the market, the model relies on two kinds of data.

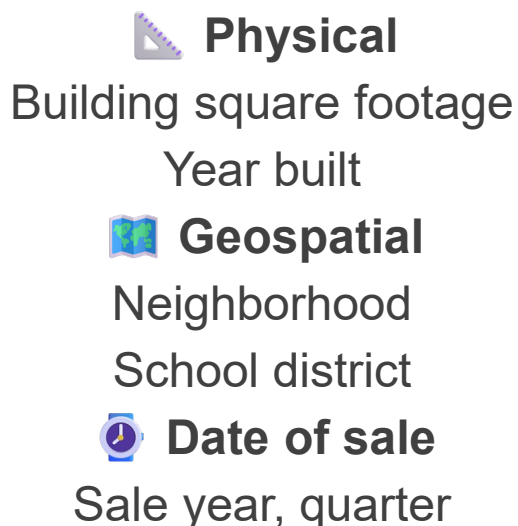
Sales

Statistical models need tens of thousands of sales to calculate price trends.



& Characteristics

Features of each home and sale *may* be predictive of sale prices.



Estimated Value

The model predicts the value of every home being reassessed.



“Okay, so...how did the model assess my home?”

Transparency, Phase 1

27 pages of **model documentation** on [github](#).
(And the actual code!)

Here is how to **search** sales on Open Data. You'll need to **filter** by PIN, or class code.

We publish **reporting dashboards** containing aggregate stats about median sale prices in your neighborhood.

Model Selection

We use [LightGBM](#) for our primary valuation model. LightGBM is a [GBDT \(gradient-boosting decision tree\)](#) framework created and maintained by Microsoft. It has [an excellent R API](#) and has been around since 2016.

We [tried a number of other model types and frameworks](#), including regularized linear models, [XGBoost](#), [CatBoost](#), random forest, shallow neural networks, and support vector machines. We even tried ensemble methods such as [model stacking](#). We chose LightGBM because it has the right mix of

← Back to Primer → Switch to Grid View

pin	#	year
33050241015		2006
23000631367		2004
01010381325		2022

1 of 21522 >

Filters | ✕ Clear all ↶ Undo

Select a column to filter...



“How did the model assess my home?”

Transparency, Phase 2: the Home Value Report



Home Alone House, in Winnetka
05-17-417-020-0000

1. Go to cookcountyassessoril.gov
2. Under the **Data** tab,
select **Home Value Report**
3. PIN: **05-17-417-020-0000**
Reassessment Year: **2025**

Part 2:

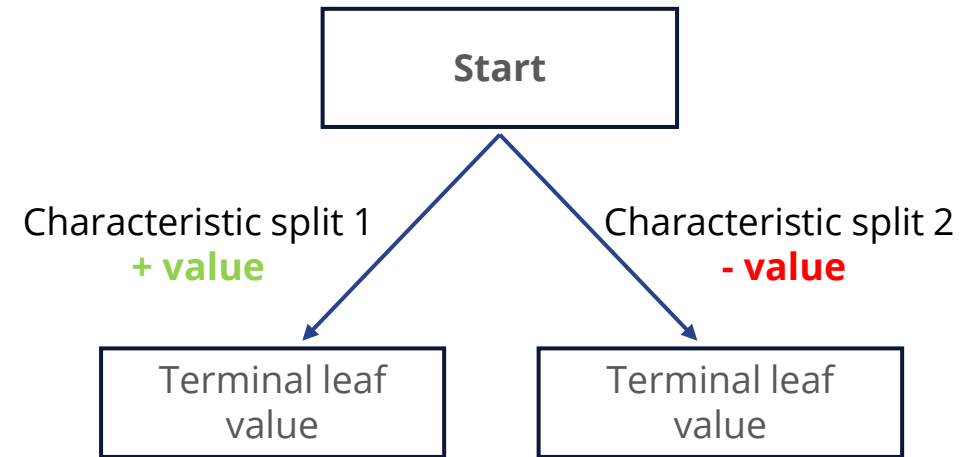
The significant sales algorithm

Powering Homeval and prioritizing
model explainability

Setting the stage

Tree models 101

- **First:** Brief explanation of tree models
- **Then:** Explain terminal leaf node matching
- **Finally:** How the algorithm selects the 5 most significant sales



HomeVal Algorithm

Tree models 101 – like a flow chart



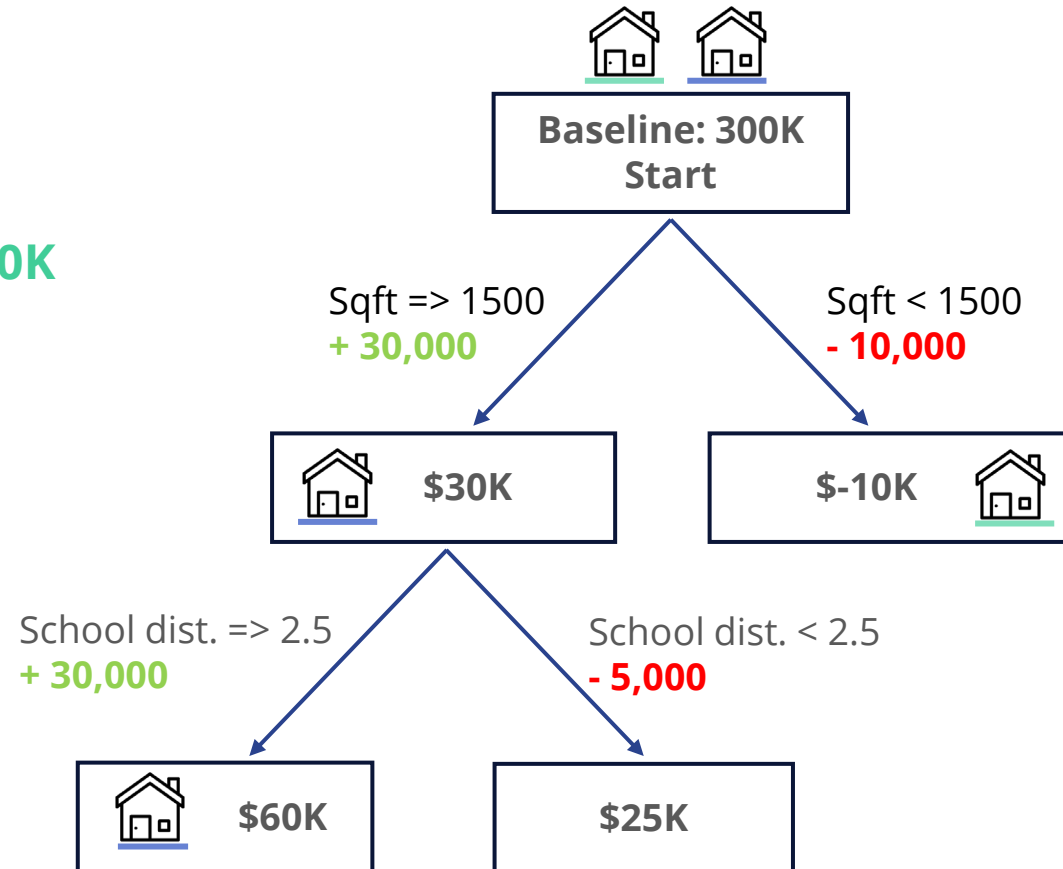
Home 1

- Sqft: 1000
- School district rating: 3
- Prediction: $\$300k - \$10K = \$290K$



Home 2

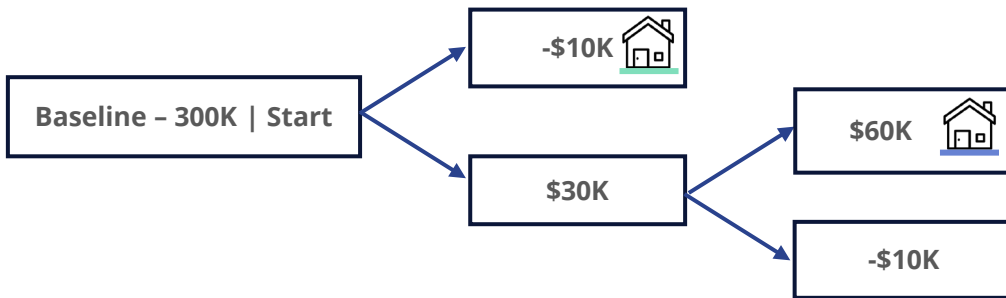
- Sqft: 2000
- School district rating: 5
- Prediction: $\$300k + \$60K = \$360K$



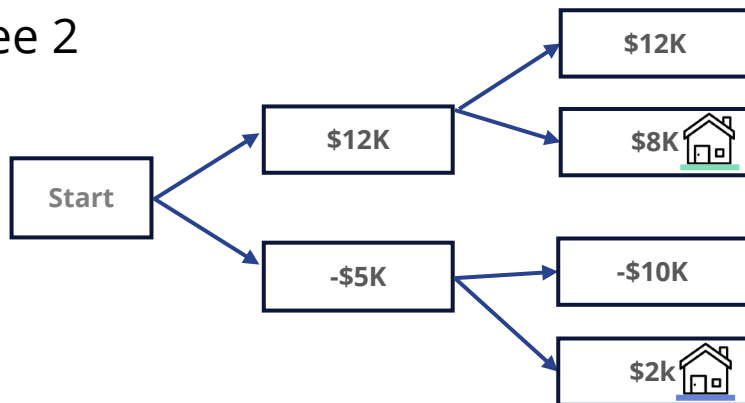
HomeVal Algorithm

Tree models 101, multiple trees

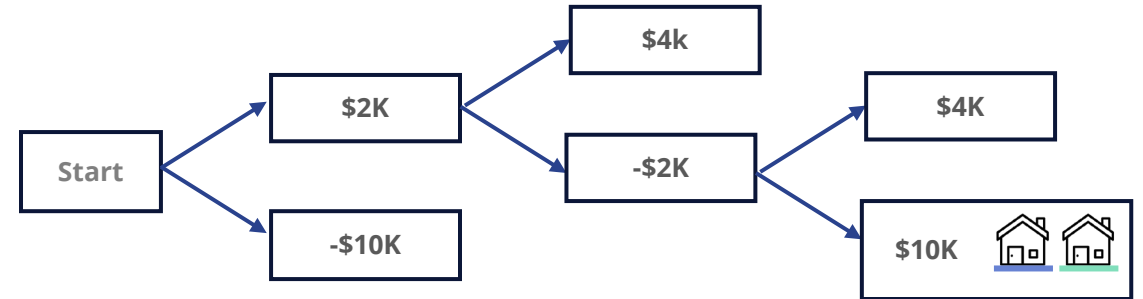
Tree 1



Tree 2



Tree 3



Home 1

- Prediction: $\$300K - \$10K + \$8K + \$10K = \text{\textcolor{teal}{\$308K}}$



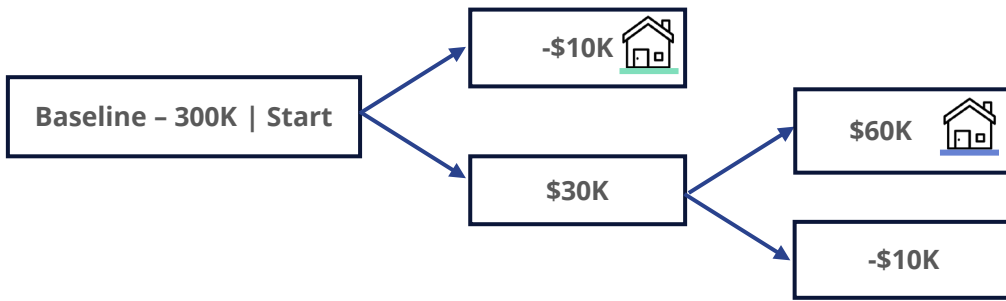
Home 2

- Prediction: $\$300K + \$60K + \$2K + \$10K = \text{\textcolor{blue}{\$372K}}$

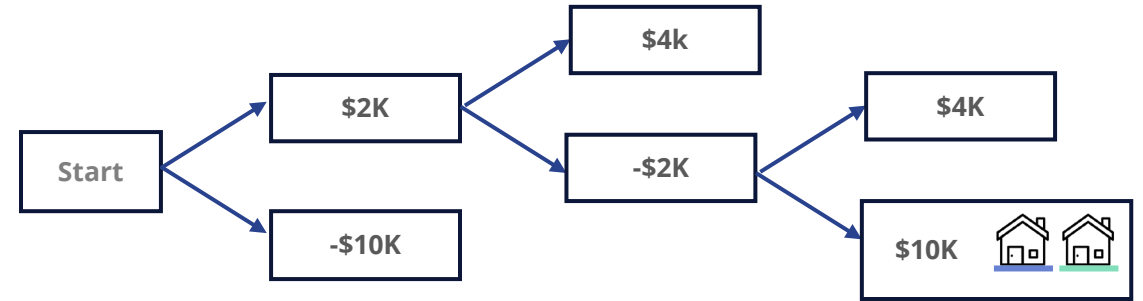
HomeVal Algorithm

Tree models 101, multiple trees

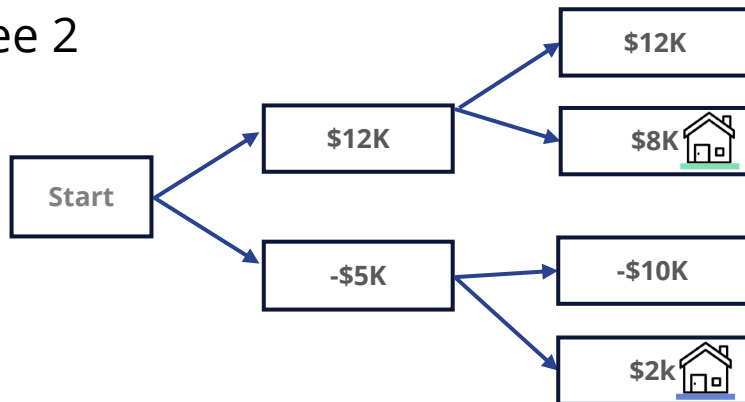
Tree 1



Tree 3



Tree 2



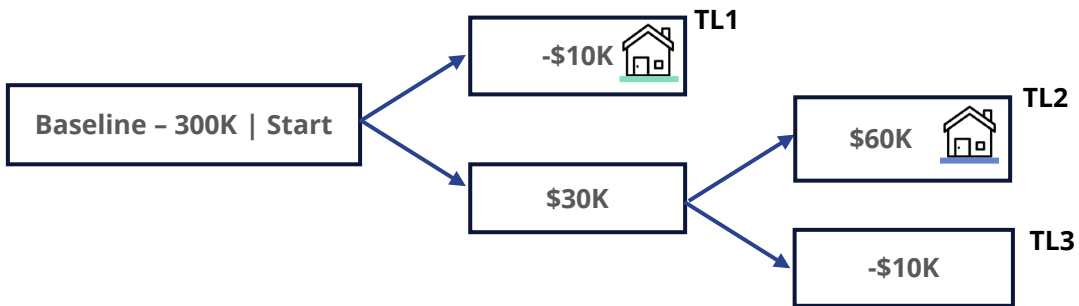
The model we use has:

- ~ 100 characteristics instead of 2, and
- over 1000 trees, with each tree having roughly 10 levels of splits and leaves

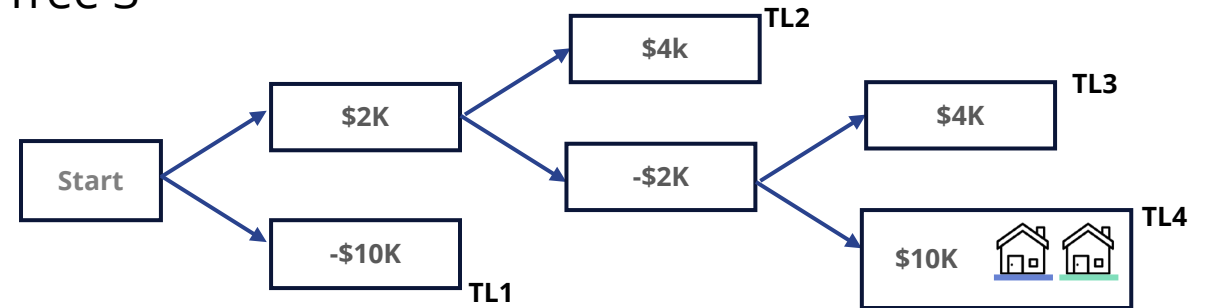
Shared terminal leaves

Setting up for similarity

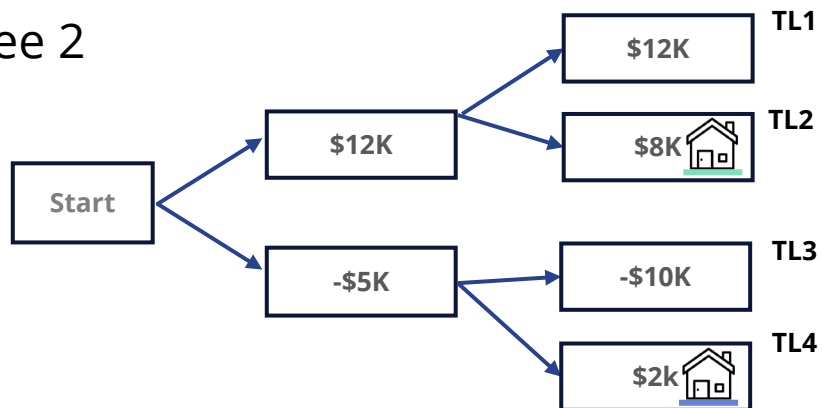
Tree 1



Tree 3



Tree 2



	Tree 1	Tree 2	Tree 3
Home 1	TL1	TL2	TL4
Home 2	TL2	TL4	TL4

Terminal leaf = TL

Finding significant sales

Comparing your home to 100,000+ sales

- We compare your home to thousands of sales
- Shared leaf nodes approximates home characteristics and the model output

	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Shared
Your home	TL1	TL4	TL2	TL1	TL3	
Sale 1	TL1 ✓	TL3 ✗	TL2 ✓	TL2 ✗	TL5 ✗	2
Sale 2	TL2 ✗	TL2 ✗	TL1 ✗	TL4 ✗	TL2 ✗	0
Sale 3	TL3 ✗	TL4 ✓	TL5 ✗	TL2 ✗	TL1 ✗	1
Sale 4	TL1 ✓	TL3 ✗	TL3 ✗	TL1 ✓	TL3 ✓	3
Sale 5	TL1 ✓	TL4 ✓	TL3 ✗	TL1 ✓	TL3 ✓	4
Sale 6	TL4 ✗	TL2 ✗	TL5 ✗	TL3 ✗	TL1 ✗	0
Sale 7	TL1 ✓	TL4 ✓	TL2 ✓	TL1 ✓	TL2 ✗	4

Top 5 most significant sales

Using tree structure to choose significant sales

- To find the top 5 most important sales, we rank them by terminal leaves shared with your home
- More on this type of algorithm in the appendix

	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Shared
Your home	TL1	TL4	TL2	TL1	TL3	
Sale 7	TL1 ✓	TL4 ✓	TL2 ✓	TL1 ✓	TL2 ✗	4
Sale 5	TL1 ✓	TL4 ✓	TL3 ✗	TL1 ✓	TL3 ✓	4
Sale 4	TL1 ✓	TL3 ✗	TL3 ✗	TL1 ✓	TL3 ✓	3
Sale 1	TL1 ✓	TL3 ✗	TL2 ✓	TL2 ✗	TL5 ✗	2
Sale 3	TL3 ✗	TL4 ✓	TL5 ✗	TL2 ✗	TL1 ✗	1
Sale 6	TL4 ✗	TL2 ✗	TL5 ✗	TL3 ✗	TL1 ✗	0
Sale 2	TL2 ✗	TL2 ✗	TL1 ✗	TL4 ✗	TL2 ✗	0

Computationally expensive

How we optimize the algorithm

Numba high performance python code



Only make comparisons for homes up for reassessment

Parallelize on multiple cores

Run on AWS to utilize more powerful machines



Part 3:

Building the app

OR: How to serve 1.9m unique pages with as little maintenance effort as possible

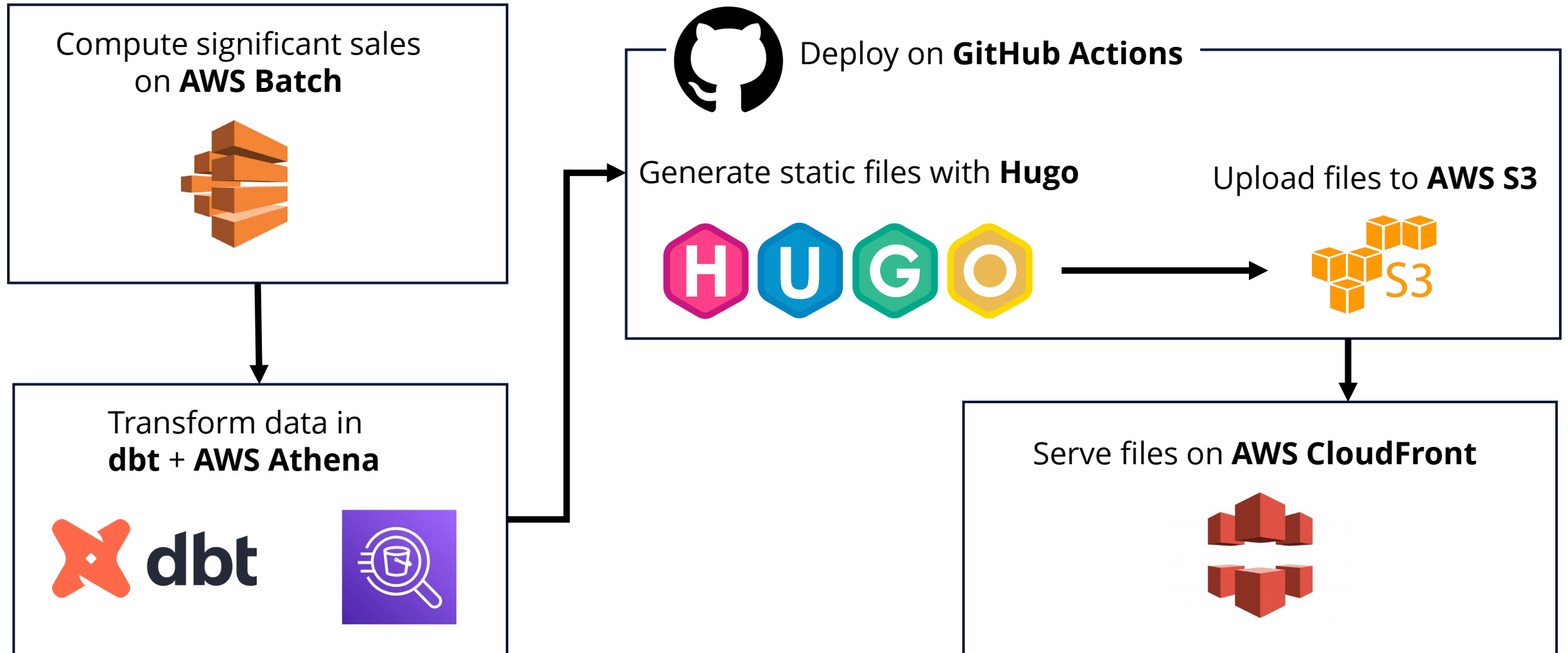
Building the app

Product goals

-  Cheap to host
-  Easy to maintain
-  Quick and simple to deploy
-  Looks beautiful
-  Mobile responsive
-  Fast response times
-  Built to last
-  Simple to understand

Building the app

Architecture diagram



Building the app

Compute significant sales on AWS Batch

☰ [CloudWatch](#) > [Log groups](#) > [/aws/batch/job](#) > [z_ci_390-test-comparables-algorithm-weighting-specification_ccao-data-model-...](#)

Log events



Actions ▼

Start tailing

Create metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

🔍 Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone ▼

Display ▼



Timestamp

Message



2025-09-19T22:04:36.537Z

Extracting weights from training data



2025-09-19T22:06:15.197Z

Calling out to python/comps.py to perform comps calculation



2025-09-19T22:06:17.279Z

Getting top 5 comps for chunk 0/10 - 29610/296091 observations, 381763 possible comps



2025-09-20T00:02:07.705Z

Getting top 5 comps for chunk 1/10 - 29609/296091 observations, 381763 possible comps



2025-09-20T01:18:45.672Z

Getting top 5 comps for chunk 2/10 - 29609/296091 observations, 381763 possible comps



2025-09-20T02:35:51.104Z

Getting top 5 comps for chunk 3/10 - 29609/296091 observations, 381763 possible comps



2025-09-20T03:53:21.850Z

Getting top 5 comps for chunk 4/10 - 29609/296091 observations, 381763 possible comps



2025-09-20T05:11:14.234Z

Getting top 5 comps for chunk 5/10 - 29609/296091 observations, 381763 possible comps

Back to top ^

Building the app

Transform data in dbt and AWS Athena

← Files master ▾ data-architecture / dbt / models / pinval / pinval.vw_assessment_card.sql ↑ Top

Code Blame Raw Copy Download Edit ▾

```
-- Get some metadata for the model runs that we want to use as the basis for
-- PINVAL reports
runs_to_include AS (
  SELECT
    model_run.run_id,
    model_run.model_predictor_all_name,
    model_run.assessment_year,
    model_run.assessment_data_year,
    model_run.assessment_triad,
    SUBSTRING(final.run_id, 1, 10) AS final_model_run_date,
    final.township_code_coverage
  FROM {{ source('model', 'metadata') }} AS model_run
  INNER JOIN eligible_card_runs AS card_runs
    ON model_run.run_id = card_runs.run_id
  INNER JOIN {{ ref('model.final_model') }} AS final
    ON model_run.run_id = final.run_id
),
```

Amazon Athena > Query editor ⓘ

Workgroup primary ▾

< Editor Recent queries Saved queries >

Query 152 : (+) ▾

```
1 SELECT *
2 FROM pinval.vw_assessment_card
3 WHERE assessment_year = '2025'
4 AND meta_township_code = '72';
```

SQL Ln 4, Col 31 ⌵ ⌴ ⚙

Building the app

Generate static files with Hugo on GitHub Actions

Summary

Jobs

- ✓ setup
- ✓ generate-pinval (10)
- ✓ generate-pinval (16)
- ✓ generate-pinval (17)
- ✓ generate-pinval (18)
- ✓ generate-pinval (20)
- ✓ generate-pinval (22)
- ✓ generate-pinval (23)
- ✓ generate-pinval (24)
- ✓ generate-pinval (25)
- ✓ generate-pinval (26)
- ✓ generate-pinval (29)
- ✓ generate-pinval (25)

generate-pinval (10)

succeeded 3 weeks ago in 16m 48s

Search logs



Generate Hugo content files and build reports

7m 15s

```
1  ▶ Run read -ra PINS <<< ""
42 Querying data from Athena ...
43 Shape of df_assessment_all: (5559, 244)
44 Computing SHAP weights for feature ranking
45 Formatting assessment dataframe for display
46 Executing comps query ...
47 Comps query finished in 134.99s
48 Shape of df_comps_all: (27795, 129)
49 Grouping by PIN took 1.45 seconds
50 Iterating pins to generate frontmatter
51 Processing PIN 1 of 5482
52 Processing PIN 5001 of 5482
53 ✓ Completed generating frontmatter for 5482 PINs in 78.7817 seconds.
54 Building Hugo reports by neighborhood ...
55 Moving /home/runner/work/pinval/pinval/hugo/temp-pinval-reports/10014 to /home/runner/work/pinval/
   pinval/hugo/content/10014 and building reports
56 Start building sites ...
57 hugo v0.147.5-7766fc62416ff95ed7768c19947ac0fcd96444cc+extended linux/amd64
```


Building the app

Upload files to AWS S3 on GitHub Actions

Amazon S3 > Buckets > ccao-pinval-prod-us-east-1 > 2025/



Objects (999+)



Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

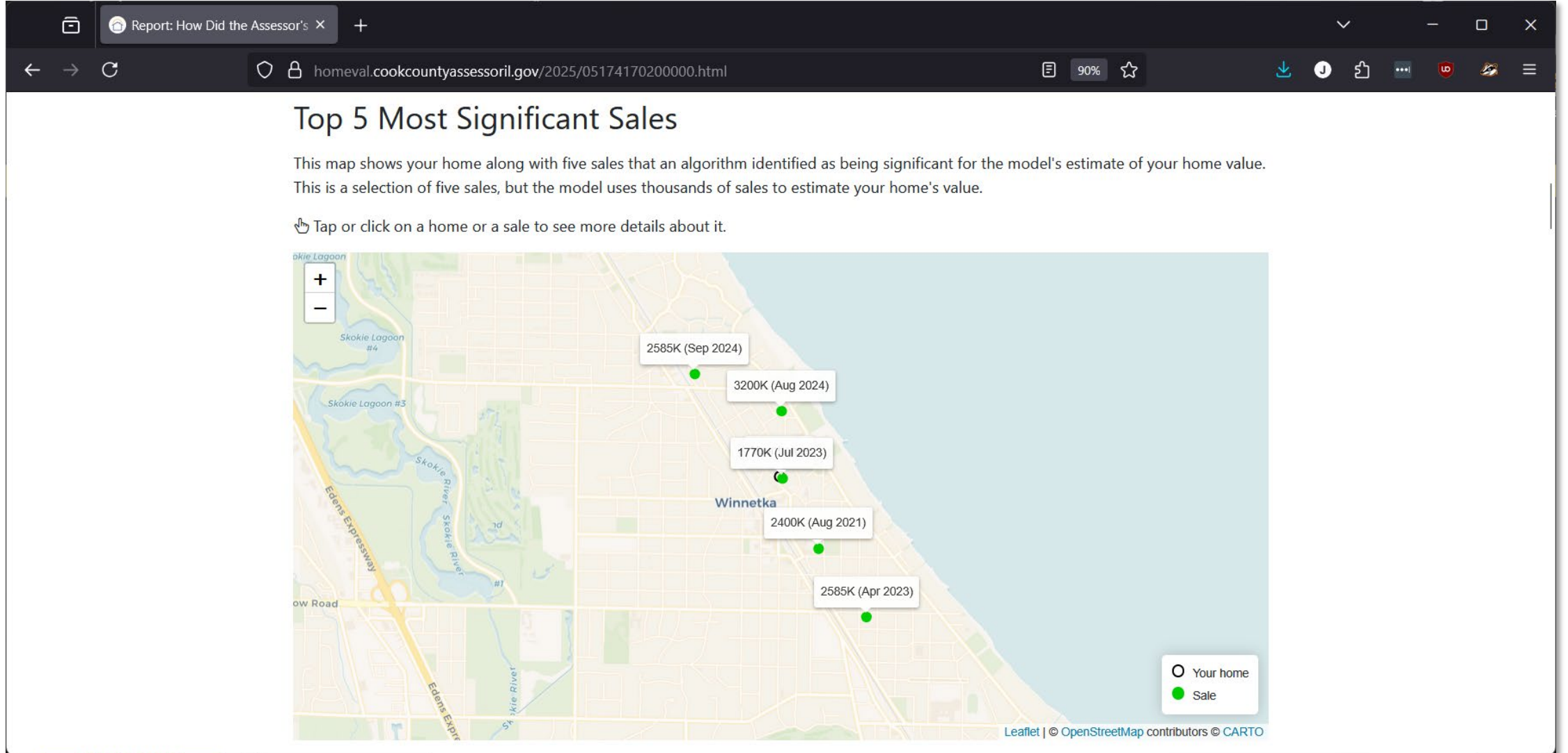
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 2 3 4 5 6 7 ... > Settings

☐	Name	Type	Last modified	Size	Storage class
☐	 01011000020000.html	html	August 27, 2025, 17:08:13 (UTC-05:00)	3.9 KB	Standard
☐	 01011000030000.html	html	August 27, 2025, 17:08:13 (UTC-05:00)	3.9 KB	Standard
☐	 01011000040000.html	html	September 11, 2025, 11:23:22 (UTC-05:00)	111.9 KB	Standard
☐	 01011000050000.html	html	September 11, 2025, 11:23:22 (UTC-05:00)	111.9 KB	Standard
☐	 01011000060000.html	html	September 11, 2025, 11:23:22 (UTC-05:00)	112.4 KB	Standard
☐	 01011000080000.html	html	August 27, 2025, 17:08:13 (UTC-05:00)	3.8 KB	Standard
☐	 01011000090000.html	html	September 11, 2025, 11:23:22 (UTC-05:00)	112.0 KB	Standard

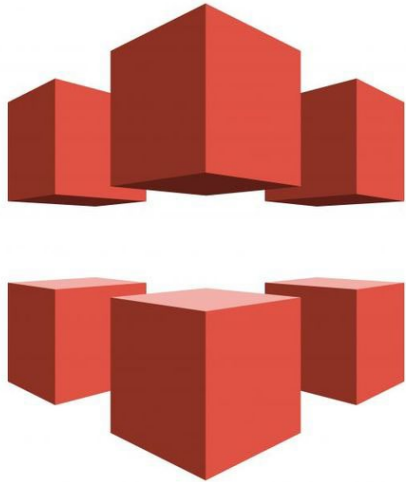
Serve files on AWS CloudFront



Building the app

Architecture tradeoffs

Architecture: **Static files hosted on S3 + CloudFront CDN**



✓ Pros

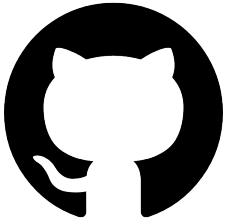
- Easy to **automate deployments**
- HTML files = complete **design flexibility**
- Extremely fast **response times**

✗ Cons

- Pay by usage = **higher hosting cost**
- Maintenance requires **AWS skills**
- Complex **deployment stack**

Thank you!

Nicole Jardine, Michael Wagner, & Jean Cochrane of the Cook County Assessor's Data Team



GitHub repo: [ccao-data/homeval](https://github.com/ccao-data/homeval)



Web app: <https://www.cookcountyassessoril.gov/home-value-report>



Algorithm: ccao-data.github.io/lightsnip/articles/finding-comps.html



Email us: **Assessor.Data@cookcountyil.gov**

Bonus

supplemental material

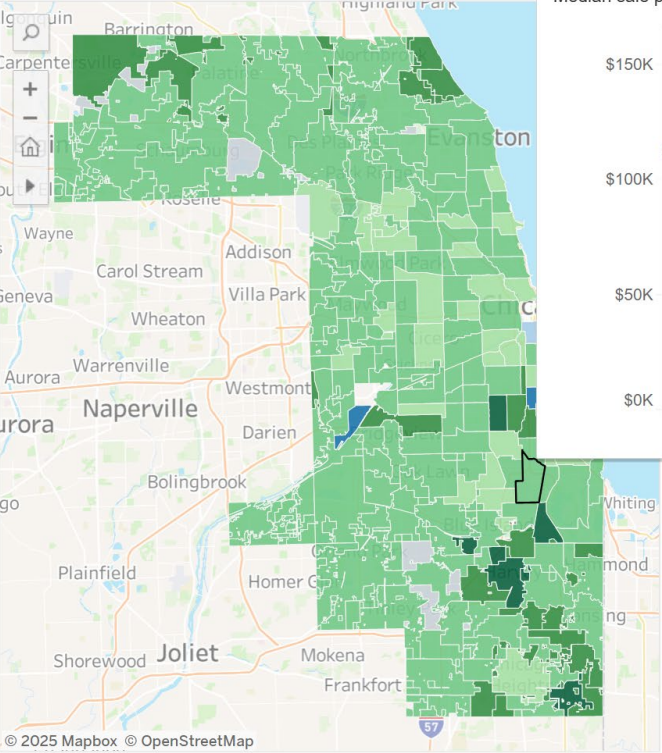
What about the other tens of thousands of sales?

cookcountyassessoril.gov/cook-county-housing-market-tracker

Housing Market Tracker

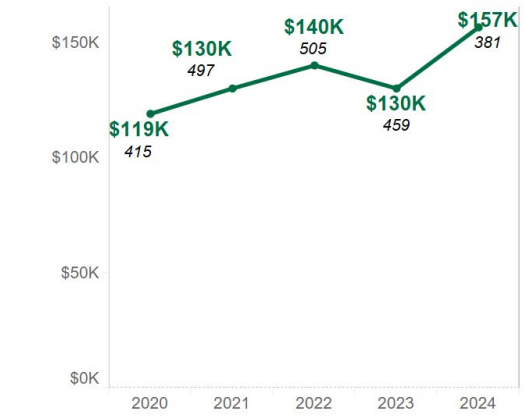
Housing prices have **gone up** in most of Cook County's suburban municipalities and Chicago's neighborhoods. Have prices in your area **gone up, gone up 100% or more**

Median Sale Price Percent Changes for **Single-Family Properties**



ROSELAND
Median Sale Price and # of Sales by year for Single-Family properties.

Median sale price increased by **32%** from 2020 to 2024.



Year	Median Sale Price	# of Sales
2020	\$119K	415
2021	\$130K	497
2022	\$140K	505
2023	\$130K	459
2024	\$157K	381

homes in Cook County.

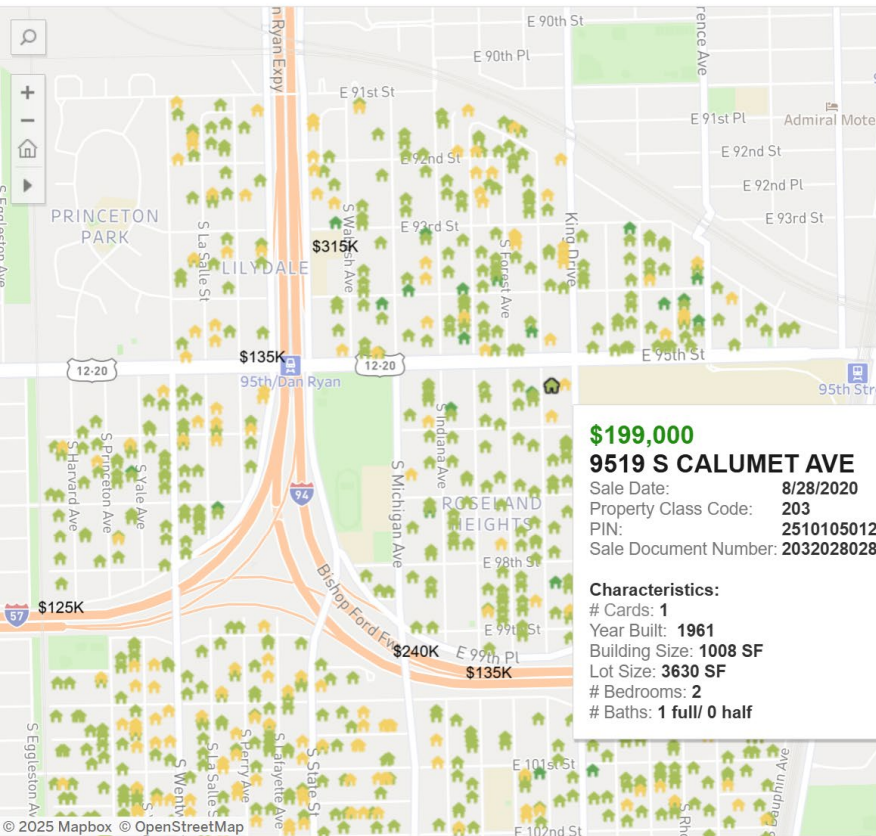
Hover over a region to see median price trends.

Click on a region to see the individual sales from 2020 to 2024.

Filter Property Type above to change the residential property group being analyzed.

ROSELAND

Individual Sales from 1/1/2020 through 12/31/2024



Back to County Map

Sale Price Bin (Select to Highlight Sales)

- \$10K - \$100K
- \$101K - \$300K
- \$301K - \$600K

Select Residential Group

Single-Family

Filter Class Codes

(All)

Filter Sale Years

(All)

Sale Price Range

\$440K

+ More Filters

map displays the most sale for a PIN or condo g. To see information on is since 2020 for a multi-family PIN or building, select an icon map to navigate to "All Recent Sales"

\$199,000

9519 S CALUMET AVE

Sale Date: 8/28/2020

Property Class Code: 203

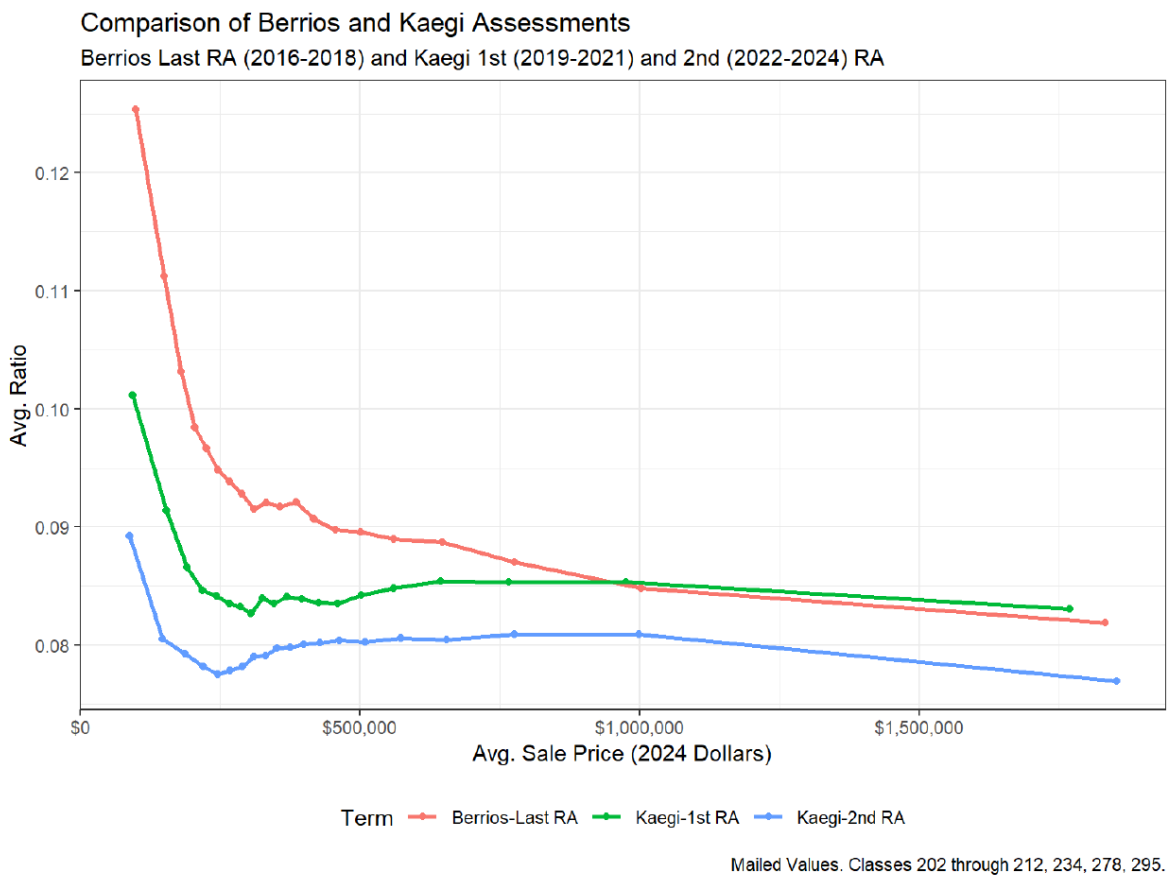
PIN: 25101050120000

Sale Document Number: 2032028028

Characteristics:

- # Cards: 1
- Year Built: 1961
- Building Size: 1008 SF
- Lot Size: 3630 SF
- # Bedrooms: 2
- # Baths: 1 full/ 0 half

Assessment improvements



[An Evaluation of Progress on Residential Assessment Fairness in Cook County](#)

This report evaluates the changes in property assessments made during Fritz Kaegi's first six years as Cook County Assessor, 2019 to 2024. The central finding is that the Kaegi administration has made substantial progress in improving the fairness of residential assessments.

Features (Not Exhaustive)

✓ = New / added feature
⚡ = Important feature

Characteristics

- Central heat and A/C
- Attic, basement chars.
- ⚡ Square footage
- ⚡ # of bed, bathrooms
- # of fireplaces
- Garage chars.
- Porch chars.
- Exterior wall type
- Fireplaces
- ⚡ Year built
- ✓ Recent renovation
- Roof type
- Property type
- Property use (SF / MF)

Geospatial

- Floodplain (FEMA)
- ✓ Flood risk (First Street)
- ✓ Airport noise
- ✓ Proximity to amenities
- ✓ Proximity to disamenities
- ⚡ School districts (also new!)
- ✓ School ratings
- ✓ Walkability
- ⚡ Assessor neighborhood

Economic

- ✓ Median income
- ✓ Foreclosures
- ✓ House price indices

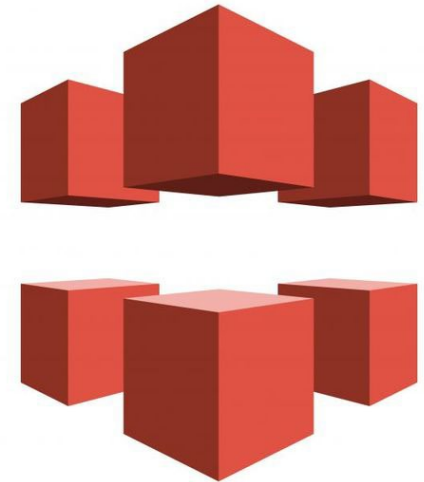
Time

- Sale day (since 1997)
- Sale year
- ✓ Seasonality features
- ✓ Sale after COVID
- ✓ Other trend features

Building the app

Deployment options

- Option 1: **Shiny app (Quarto) running on a server**
- Option 2: **Tableau dashboard**
- Option 3: **Static files on S3 + CloudFront CDN**



Building the app

Deployment options

- Option 1: **Shiny app (Quarto) running on a server**



-  **Pros**
 - Widespread **R knowledge** on our team
 - On-prem deployment = **no hosting fees**
 - Prototype **already built** in Quarto
-  **Cons**
 - Slow **response times**
 - Can't **scale** easily in response to increased load
 - Maintenance requires **Linux sysadmin skills**

Building the app

Deployment options

- Option 2: **Tableau dashboard**



- **Pros**

- Widespread **Tableau knowledge** on our team
- Existing license = **no extra hosting fees**
- Prototype **already built**

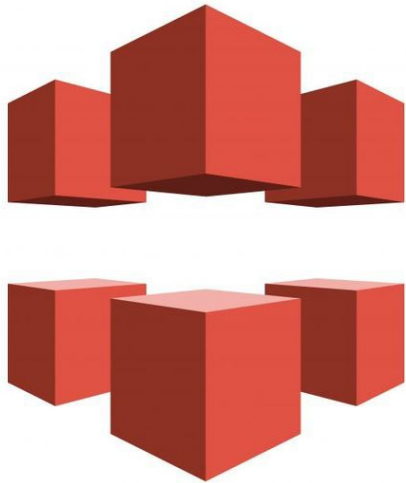
- **Cons**

- Slow **response times**
- No **git version control**
- Difficult to make **mobile responsive**

Building the app

Deployment options

- Option 3: **Static files on S3 + CloudFront CDN**



-  **Pros**
 - Easy to **automate deployments**
 - HTML files = complete **design flexibility**
 - Extremely fast **response times**
-  **Cons**
 - Pay by usage = **highest hosting cost**
 - Maintenance requires **AWS skills**
 - Complex **deployment stack**

Building the app

Deployment options

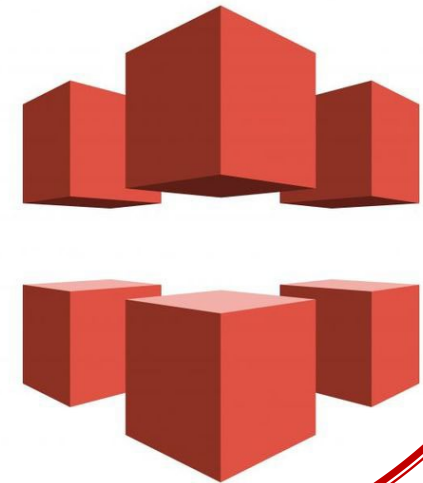
- Option 1: **Shiny app (Quarto) running on a server**



- Option 2: **Tableau dashboard**



- Option 3: **Static files on S3 + CloudFront CDN**



Winner

How characteristic scores work

Overview of characteristic scores

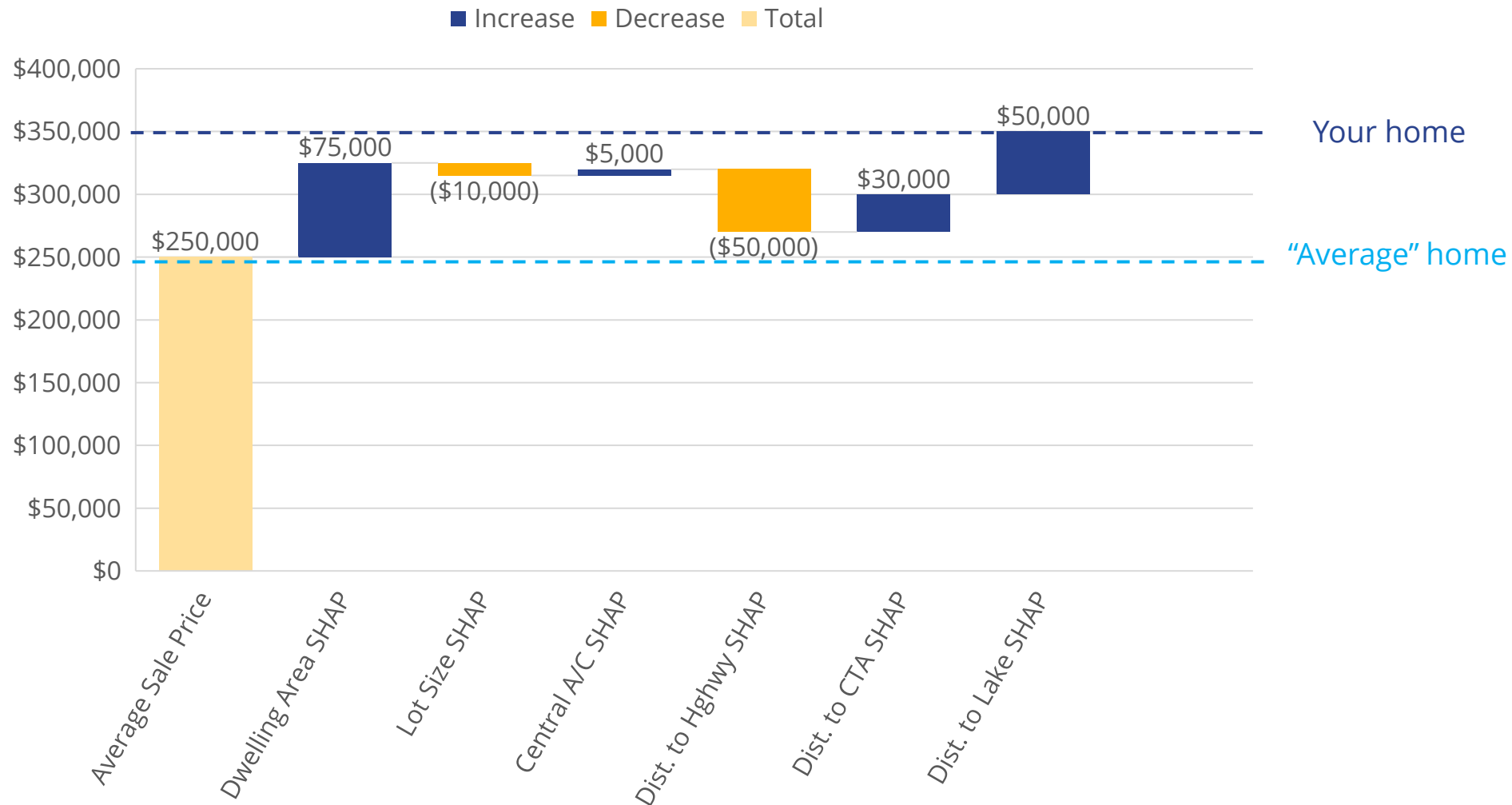
Characteristic	Score 	Your home	Sale 1	Sale 2	Sale 3	Sale 4	Sale 5
PIN		05-17-417-020	05-21-109-009	05-17-404-033	05-17-417-021	05-17-202-003	05-21-303-013
<u>Neighborhood Code</u>	★★★★	23022	23022	23022	23022	23021	23022
<u>Property Class</u>	★★★★	209	209	209	206	209	208
<u>Year Built</u>	★★★	1921	1920	1977	1914	1939	2007
<u>Building Square Feet</u>	★★★★	5,083	5,392	5,588	4,107	5,310	4,101
<u>Land Square Feet</u>	★★★★	23,278	15,500	17,773	13,222	22,764	14,390
<u>Bedrooms</u>	★★	6	6	4	6	8	6
<u>Full Baths</u>	★★★★	5	3	4	3	4	5
<u>Half Baths</u>	★★	1	1	1	1	0	1
<u>Township Code</u>	★★	23	23	23	23	23	23

How characteristic scores work

SHAP values measure how each characteristic distinguishes a home from the “average” home

Hypothetical home characteristics:

- Large dwelling area
- Small lot
- Central A/C
- Close to highway
- Close to CTA
- Close to Lake Michigan



How characteristic scores work

SHAP values add up to the predicted value

For every parcel with N characteristics:

$$\begin{aligned} \text{Prediction} &= \text{Median Sale Price} + \text{Characteristic}_1\text{SHAP} + \text{Characteristic}_2\text{SHAP} + \dots + \text{Characteristic}_n\text{SHAP} \\ &= \text{Median Sale Price} + \text{Building S.F. SHAP} + \text{Land S.F. SHAP} + \dots \text{School District SHAP} \end{aligned}$$

How characteristic scores work

SHAP values measure how each characteristic distinguishes a home from the “average” home

For every parcel with N characteristics:

$$\begin{aligned} \text{Prediction} &= \text{Average Sale Price} + \text{Characteristic}_1\text{SHAP} + \text{Characteristic}_2\text{SHAP} + \cdots + \text{Characteristic}_n\text{SHAP} \\ &= \text{Average Sale Price} + \text{Building S.F. SHAP} + \text{Land S.F. SHAP} + \cdots + \text{Dist. to Lake SHAP} \end{aligned}$$

How characteristic scores work

SHAP values measure how each characteristic distinguishes a home from the “average” home

Given:

Median Sale Price = \$250K

Building S.F. SHAP = \$20K

Land S.F. SHAP = -\$5K

Dist. to Lake SHAP = \$10K

Then:

Prediction = *Average Sale Price* + *Building S.F. SHAP* + *Land S.F. SHAP* + *Dist. to Lake SHAP*

= \$250K + \$20K - \$5K + \$10K

= \$275K

How characteristic scores work

Characteristic scores compare the relative size of SHAP values

For every characteristic in a given parcel:

$$char\ score_i = \frac{| char_i SHAP |}{\sum_{k=1}^n | char_k SHAP |}$$

How characteristic scores work

Characteristic scores compare the relative size of SHAP values

Given:

Median Sale Price = \$250K

Building S.F. SHAP = \$20K

Land S.F. SHAP = -\$5K

Dist. to Lake SHAP = \$10K

Then:

$$\begin{aligned} \text{char score}_{\text{Building S.F.}} &= \frac{|\text{Building S.F. SHAP}|}{\sum_{k=1}^n |\text{char}_k \text{ SHAP}|} \\ &= \frac{|\$20K|}{|\$20K| + |-\$5K| + |\$10K|} = \frac{\$20K}{\$35K} \cong 0.57 \end{aligned}$$

How characteristic scores work

Simplify characteristic scores by binning them

★	→	$\text{char score} < 0.0005$
★★	→	$0.0005 \leq \text{char score} < 0.005$
★★★	→	$0.005 \leq \text{char score} < 0.025$
★★★★	→	$0.025 \leq \text{char score}$

Significant sales algorithm

More information

- The creators of the random forest algorithm call this system of terminal leaf node matching ‘proximities’

Breiman, L. Random Forests. *Machine Learning* 45, 5–32 (2001).
<https://doi.org/10.1023/A:1010933404324>

https://www.stat.berkeley.edu/~breiman/RandomForests/cc_home.htm

RANDOM FORESTS

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January 2001

Proximities

These are one of the most useful tools in random forests. The proximities originally formed a $N \times N$ matrix. After a tree is grown, put all of the data, both training and oob, down the tree. If cases k and n are in the same terminal node increase their proximity by one. At the end, normalize the proximities by dividing by the number of trees.

Significant sales algorithm

Different variations

Our implementation of this algorithm is based on equally weighting trees based on whether terminal leaf nodes match

However, there are different specifications of this algorithm that can augment the logic to weight each tree's shared terminal leaf node contribution to an overall score

Significant sales algorithm

Other variations – weighting by error reduction in training data

- Weight trees by how much they reduce error of the significant sale in training data

ccao-data.github.io/lightsnip/articles/finding-comps.html

Finding comparables with LightGBM

Source: `vignettes/finding-comps.rmd`

The [Cook County Assessor's Office \(CCAO\)](#) uses LightGBM for its [residential](#) and [condominium](#) valuation models. [LightGBM](#) is a machine learning framework that works by iteratively growing many [decision trees](#). It is especially suited to data with many categorical variables and complex interactions, such as housing data.

However, LightGBM is also complicated. Its outputs can be difficult to explain and interpret, especially as model complexity grows. Techniques such as [SHAP values](#) can help, but aren't intuitive to the average person.

This vignette outlines a novel technique to explain LightGBM outputs specifically for housing data. The technique finds [comparable sales](#) from the model training data by exploiting the tree structure of a LightGBM model. Its goal is to help diagnose model issues and answer a common question from property owners, "What comparable sales did you use to value my property?"

ID	Tree 0	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Tree 6	Tree 7	Tree 8	Tree 9	Sim. Score
3	71.2%	4.7%	4.2%	3.8%	3.4%	3.1%	0.0%	0.0%	0.0%	0.0%	90.46%
14	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
21	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
48	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
55	71.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	71.18%
60	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
61	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
62	71.2%	4.7%	4.2%	3.8%	3.4%	3.1%	0.0%	0.0%	0.0%	0.0%	90.46%
70	71.2%	4.7%	4.2%	3.8%	3.4%	0.0%	0.0%	0.0%	0.0%	0.0%	87.37%
84	71.2%	0.0%	4.2%	3.8%	3.4%	3.1%	0.0%	0.0%	0.0%	0.0%	85.76%

Significant sales algorithm

Other variations – weighting based on variance

Tree Space Prototypes: Another Look at Making Tree Ensembles Interpretable

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DEFINITION 2. The GBT proximity of a pair of points is a weighted average of the number of trees in the GBT model in which the points end up in the same leaf:

$$\text{proximity}^{\text{GBT}}(s, s') = \sum_{i=1}^t \sum_{j=1}^{\tau_i} \frac{w_i}{\sum_{i=1}^t w_i} \mathbb{I}(s \in R_{j,i}) \mathbb{I}(s' \in R_{j,i}),$$

where the i th tree's weight w_i is

$$w_i = \gamma_i^2 \cdot \text{Var}\{c_i^{\text{Tree}}(s) : s \in S\}.$$

The GBT distance between a pair of points is then:

$$d^{\text{GBT}}(s, s') = 1 - \text{proximity}^{\text{GBT}}(s, s').$$

- Tan, S., Soloviev, M., Hooker, G., & Wells, M. T. (2020). Tree space prototypes: Another look at making tree ensembles interpretable. *Proceedings of the 2020 ACM-IMS Foundations of Data Science Conference (FODS '20)*, 23–34. <https://doi.org/10.1145/3412815.3416893>

Existing explainability tools failed internal evaluation.

Google

tree model explainability

AI Mode All Images Shopping Videos Short video



DataCamp

<https://www.datacamp.com> > ... > Artificial Intelligence

Explainable AI, LIME & SHAP for Model Interpretability

May 10, 2023 — **Explainable AI** refers to a set of processes and methods that generate human-understandable explanation for the decisions generated by machine learning models.



Medium · Pacmed

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Explainability for tree-based models: which SHAP

A widely used method to explain tree-based models is the **TreeSHAP** algorithms. In this article we have presented some ...

Top and Bottom 3 Attributes for Building #1

Rank	Category	Subcategory	Feature	Attribute	
#1	Home Characteristics	Characteristics	Building Square Feet	3889.0	\$130K
#2	Time Trending	7-Year Trends	Sale Day Since 1997	9131.0	\$36K
#3	Home Characteristics	Characteristics	Year Built	2003.0	\$30K
#92	Location Factors	Demographics of census tract	Percent Population Education, Bachelor Degree	0.15740984469597263	-\$5K
#93	Home Characteristics	Characteristics	Bedrooms	3.0	-\$6K
#94	Time Trending	7-Year Trends	Sale Day of Year	1	-\$7K

How much did # beds and baths contribute to value?

Feature	Attribute	
Full Baths	3.0	\$26K
Half Baths	0.0	-\$2K
Bedrooms	3.0	-\$6K

“How did the model assess *my* home?”

Transparency, Phase 2

Goal: A web app that explains how the model assessed a home, by showing significant sales and every characteristic used by the model.

- more **specific** to an individual home than the general Valuation Reports,
- more **accessible** than requiring the user to filter sales on a data portal,
- more **intuitive** than reading through code and documentation, and
- **immediately available** without having to go through FOIA.